

Monocular Depth Estimation as a Domain-Invariant Representation for Vision-Based Quadruped Parkour

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Abstract

Vision-based quadruped parkour typically relies on onboard depth cameras, limiting deployment to robots with specialized sensors. We investigate whether a pre-trained *monocular depth estimator* can serve as a domain-invariant bridge, enabling sim-to-real transfer from only RGB without photorealistic rendering or domain randomization.

We present a two-phase pipeline: a privileged teacher trained via PPO with ground-truth terrain scans, distilled into a visual student via DAgger operating on depth images from Depth Anything V2. In simulation, we demonstrate that this pipeline achieves limited parkour capabilities under conditions characterized by a *Depth Signal Quality* metric. Domain invariance experiments show that DA2 provides strong robustness to color and brightness changes and generalizes across simulated environments.

1 Introduction

Legged robots that traverse irregular terrain—stairs, hurdles, gaps—must perceive geometry and maintain tight coordination between vision and control. Training locomotion policies in simulation with reinforcement learning (RL) and transferring them to real hardware has proven effective for agile skills [8, 10, 14], and recent work [2, 4, 7, 22] has extended this to parkour—climbing, jumping, and crawling—via teacher-student distillation. In these systems, a *teacher* trains with privileged terrain information (e.g. ground-truth heightmaps), then a *student* reproduces the teacher’s behavior from onboard sensors.

A critical assumption, however, is access to an onboard depth sensor. Depth cameras (e.g. Intel RealSense) transfer well from simulation ??, but not all platforms include them—the Unitree Go2, for instance, ships with only a front-facing RGB camera ??.

Deploying RGB-trained policies on real robots requires bridging the *sim-to-real gap*. Three strategies have

emerged: (1) **domain randomization** [18], which randomizes visual properties; (2) **generative augmentation** [21], which uses diffusion models for photorealistic training images; and (3) **canonical representations**, which project both domains into a shared space via a foundation model.

We pursue strategy (3) using Depth Anything V2 (DA2) [20], a pre-trained monocular depth estimator trained on millions of diverse real images. We propose that because DA2 maps arbitrary RGB to a depth space invariant to appearance, a student policy consuming depth should inherit that invariance without photorealistic rendering or explicit domain randomization.

Thesis statement. We hypothesize that pre-trained monocular depth estimation provides a domain-invariant intermediate representation for vision-based quadruped parkour, allowing a student trained on estimated depth in simulation to generalize across visual domains.

Contributions.

- A two-stage teacher-student pipeline from privileged scandot heightmaps to monocular depth for quadruped parkour.
- A demonstration that monocular depth enables basic parkour (stairs, hurdles, gaps) in simulation.
- Identification of depth signal quality—governed by model scale and terrain texture—as the key factor limiting performance.
- A domain invariance experiment testing DA2-based students under visual perturbations.

2 Related Work

RL for legged locomotion. Modern quadruped controllers are typically learned via reinforcement learning in simulation and transferred to real hardware [8, 14]. Chen et al. [3] introduced the privileged teacher-student paradigm, where a teacher with access to ground-truth state is distilled into a sensorimotor student. Lee et al. [10] applied this to locomotion, training a student from proprioceptive history alone and achieving zero-shot sim-to-real transfer on rough terrain.

Miki et al. [11] extended this framework to perceptive locomotion, incorporating exteroceptive depth inputs.

Vision-based parkour. Agarwal et al. [1] introduced the scandot representation for egocentric depth-based locomotion, distilling a privileged teacher into a visual student. Extreme Parkour [4] and Robot Parkour Learning [22] build on this with CNN-GRU students that process depth images, demonstrating climbing, jumping, and crawling on real quadrupeds. SoloParkour [2] replaces distillation with off-policy RL warm-started from privileged experience, and ANYmal Parkour [7] uses a hierarchical skill-selection policy. All four systems assume access to a depth camera at deployment. Jenkins [9] is the closest work to ours, training an RGB parkour policy via domain randomization; we instead use monocular depth estimation as a canonical intermediate representation.

Sim-to-real for vision. Domain randomization [18] perturbs visual properties (textures, lighting, colors) so the policy learns invariance through exposure. LucidSim [21] replaces randomization with generative models that produce photorealistic training images. Xue et al. [19] combine photorealistic simulation with a teacher-student-bootstrap framework for humanoid loco-manipulation from RGB. All three approaches require significant rendering or generation overhead; we sidestep this by projecting RGB into a canonical depth space via a pre-trained estimator.

Monocular depth estimation. MiDaS [12] pioneered monocular depth estimation across diverse datasets. Depth Anything V2 (DA2) [20] achieves state-of-the-art zero-shot depth estimation using a vision transformer trained on 142M images. DA2 offers both relative and metric variants; we use the metric model, which predicts absolute depth in meters. Figure 1 illustrates the qualitative behavior of DA2 across diverse scenes, which is the motivation for using it here as a canonical visual representation.



Figure 1: Depth Anything V2 demonstration [5, 20].

3 Mathematical Foundations

3.1 Problem Formulation

We model the parkour task as a Partially Observable Markov Decision Process $(\mathcal{S}, \mathcal{A}, \mathcal{O}, T, R, \gamma)$ with state space $\mathcal{S}$, action space $\mathcal{A}$, observation space $\mathcal{O}$, transition dynamics T , reward function R , and discount factor γ . The agent receives partial observation $o_t \in \mathcal{O}$ and

selects action $a_t \in \mathcal{A} = \mathbb{R}^{12}$ (target joint position offsets). Observations decompose into three groups:

- **Proprioception** $o_{\text{prop}} \in \mathbb{R}^{53}$: base angular velocity, projected gravity, velocity and heading commands, joint positions, joint velocities, previous actions, and foot contacts.
- **Exteroception** o_{ext} : scandot heightmaps ($\mathbb{R}^{132}$) for the teacher, or depth/RGB images for the student.
- **Privileged** o_{priv} : friction and other physics parameters available only during teacher training.

3.2 PPO

The teacher policy π_θ is trained with Proximal Policy Optimization (PPO) [16], which maximizes a clipped surrogate objective:

$$L^{\text{CLIP}} = \mathbb{E}_t \left[\min \left(r_t \hat{A}_t, \text{clip}(r_t, 1-\epsilon, 1+\epsilon) \hat{A}_t \right) \right], \quad (1)$$

where $r_t = \pi_\theta(a_t|o_t)/\pi_{\theta_{\text{old}}}(a_t|o_t)$ is the probability ratio and $\hat{A}_t$ is the GAE [15] advantage estimate.

3.3 Reward Function

The teacher uses a standard shaped reward that combines task progress with regularization for stable locomotion. Positive terms encourage tracking the commanded velocity and heading, making forward progress, and reaching waypoints and course completion. Penalty terms discourage early termination, collisions, unstable body motion, large torques and joint accelerations, abrupt action changes, and violations of joint or torque limits.

3.4 DAgger Distillation

The student policy is trained via Dataset Aggregation (DAgger) [13]. At each step the student observes o_{prop} and a depth image and produces a_{student} , but the teacher’s action a_{teacher} is executed. The student minimizes:

$$\mathcal{L} = \underbrace{\|a_{\text{teacher}} - a_{\text{student}}\|_2}_{\text{action loss}} + \underbrace{\|y_{\text{oracle}} - \hat{y}\|_2}_{\text{yaw loss}}, \quad (2)$$

where $\hat{y} \in \mathbb{R}^2$ is the student’s predicted heading and y_{oracle} is the ground-truth heading.

4 System Architecture

4.1 Overview

All experiments use the Genesis GPU simulator [6] with the legged_gym framework [14] and rsl_rl [17] on 4× NVIDIA L40S GPUs. The robot is a Unitree Go2 quadruped with 12 actuated joints. Actions $a_t \in \mathbb{R}^{12}$ are target joint positions, updated at 50Hz and converted to torques via a PD controller $\tau = K_p(a_t - q) - K_d\dot{q}$.

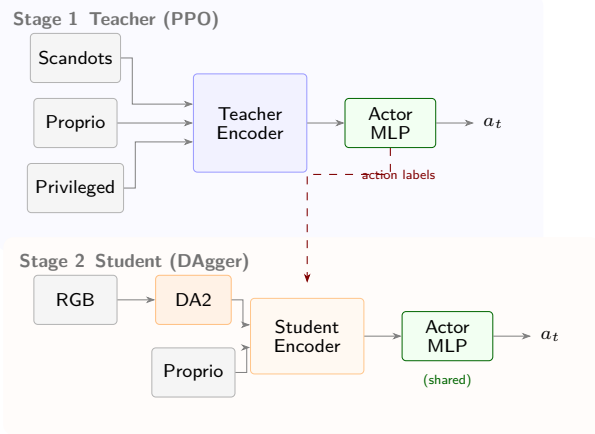


Figure 2: Two-stage training pipeline. The teacher uses privileged scandot observations; the student replaces these with either GT depth, DA2-inferred depth.

The system follows the two-stage privileged distillation pipeline described in Section 1: a teacher trains with ground-truth scandot heightmaps [1], then a student reproduces the teacher’s behavior from depth or RGB images via DAgger (Figure 2).

4.2 Teacher

The teacher policy π_T maps proprioception, privileged physics parameters, and scandot heightmaps to joint targets. Three parallel MLPs encode each observation group into 32-dim latent vectors, which are concatenated with proprioception and fed through the actor MLP $\mathbb{R}^n \rightarrow \mathbb{R}^{12}$ to produce joint position offsets.

4.3 Student

The student replaces the scandot encoder with a visual encoder that must recover the terrain latent $z \in \mathbb{R}^{32}$ from a depth image (Figure 3). A CNN maps the depth image $d \in \mathbb{R}^{58 \times 87}$ to a 32-dim feature, which is concatenated with proprioception and compressed by an MLP. A GRU integrates these fused features across timesteps, producing z and a heading estimate $\hat{y} \in \mathbb{R}^2$. The teacher’s actor MLP is reused, taking $[o_{\text{prop}}; z]$ to output joint targets. During DAgger distillation, the student minimizes the loss in Eq. 2 while the teacher’s actions are executed in the environment.

4.4 Terrain and Curriculum

The simulator generates parkour terrains with three obstacle families (stairs, hurdles, gaps) at four difficulty levels per family (Figure 4) with gap widths from 0.24 m to 0.50 m. While these obstacles are simpler than those in prior work [2, 4, 22], our goal is to establish a baseline for evaluating the DA2 depth pipeline rather than to push the limits of agile locomotion. During teacher

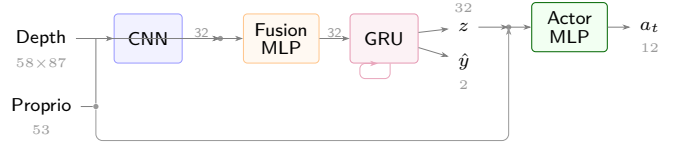


Figure 3: Student visual encoder. The depth CNN and proprioception are fused and processed by a GRU that produces the terrain latent and heading estimate used by the teacher’s actor MLP.

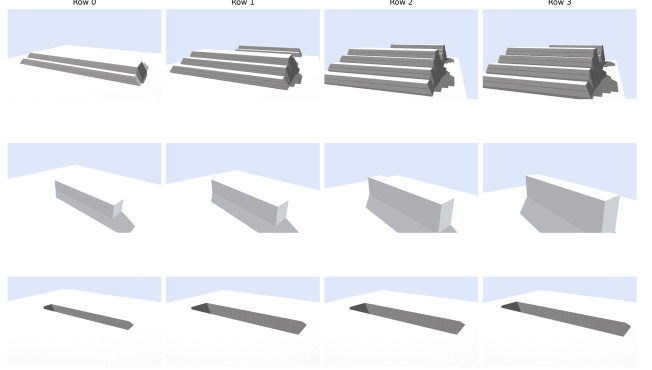


Figure 4: Parkour curriculum progression across obstacle families. Difficulty increases from left to right by increasing step count/height, hurdle size, and gap width.

training, a curriculum advances robots to harder rows upon reaching a success threshold.

5 Experiments and Results

We evaluate the DA2 pipeline through three experiments: (1) establishing upper-bound baselines with ground-truth sensing, (2) closing the gap between estimated and ground-truth depth by varying model capacity and terrain texture, and (3) testing domain invariance under visual perturbations. We report *success rate* (fraction of episodes completing the full course) and *progress ratio* (mean fractional distance along the course before termination), both averaged over 256 episodes. When analyzing visual robustness, we also report *depth signal quality* (DSQ), defined as the Pearson correlation between ground-truth and DA2-estimated depth images, averaged over environments and time steps. Pearson correlation is scale- and offset-invariant, making it appropriate because DA2 and the simulator use different depth scales:

$$\text{DSQ} = \frac{1}{T} \sum_{t=1}^T \frac{1}{N} \sum_{i=1}^N \frac{\sum_p (d_{i,p}^{\text{GT}} - \bar{d}_i^{\text{GT}})(d_{i,p}^{\text{DA2}} - \bar{d}_i^{\text{DA2}})}{\|d_i^{\text{GT}} - \bar{d}_i^{\text{GT}}\|_2 \|d_i^{\text{DA2}} - \bar{d}_i^{\text{DA2}}\|_2} \quad (3)$$

5.1 Baselines

We start by establishing upper bounds for the privileged teacher and the ground-truth depth sensing student (GT-depth). The privileged teacher achieves near-perfect success across all gap difficulties, shown in Table 1.

The GT-depth student, receiving simulator depth images, achieves success on all rows, confirming the CNN-GRU architecture extracts sufficient geometry from a single depth viewpoint. A zero-depth ablation (all depth pixels set to zero) drops success to 8%, verifying the policy relies on depth rather than exploiting proprioceptive heuristics.

Table 1: Baseline success rates on gap terrain (%).

	Row 0	Row 1	Row 2	Row 3
Scandots teacher	98.5	100.0	99.0	98.5
GT-depth student	94.5	94.5	89.1	87.5
Zero-depth ablation	8.0	2.0	0.0	0.0

5.2 Parkour from Estimated Depth

We now determine whether parkour can be achieved from estimated depth. Starting from the GT-depth setup, we replace simulator depth with DA2-small. As shown in Table 2, DA2-small achieves success on early rows, but fails completely on Rows 2 and 3. Switching to DA2-base results in promotion to Row 3, but it still falls well short of GT-depth on the hardest rows.

Figure 5 shows training dynamics, where reward rises steeply as the policies master flat terrain, then plateau as the adaptive curriculum exposes progressively harder obstacles. We show the same pattern—the textured model is able to surpass the untextured models and approach the GT-depth student’s performance.

Table 2: Policy success (%) as model scale and terrain texture are added to the estimated-depth pipeline.

Configuration	Row 0	Row 1	Row 2	Row 3
GT-depth baseline	94.5	94.5	89.1	87.5
DA2-small	56.3	33.6	—	—
DA2-base	68.4	83.2	26.6	0.8
DA2-base + texture	87.5	93.0	87.1	80.5

The remaining limitation is the visual input. On untextured terrain, the gap boundary provides weak monocular cues, so the depth estimate is ambiguous. As shown in Table 2, adding checkerboard terrain texture makes the gap geometry visible and closes most of the remaining gap to GT-depth. Figure 6 shows examples of depth estimation directly across representative surface conditions. Featureless terrain produces the weakest obstacle localization and the lowest DSQ, while checkerboard,

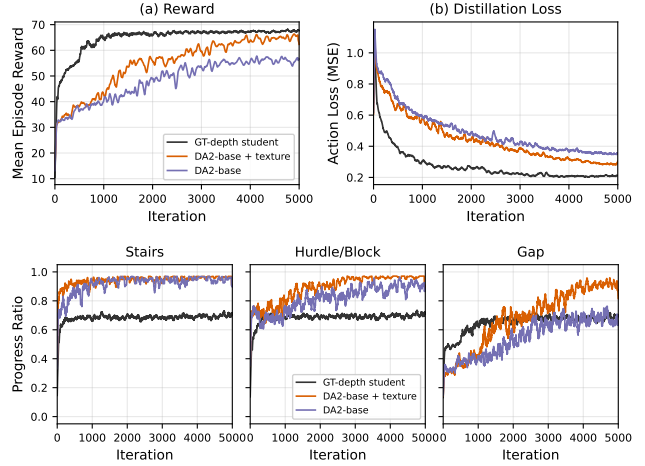


Figure 5: Success rate during student training. Breaking success down by obstacle type shows most of the gain comes from gap traversal; stairs and hurdles are much less sensitive to texture because their geometry remains visible.

grass, and wood all preserve substantially more geometric structure.

5.3 Domain Invariance

We now test whether DA2 provides invariance to the visual domain shifts encountered in sim-to-real transfer. We apply perturbations at *test time only* to the trained DA2-base+texture student, spanning texture replacement, lighting and color shifts, and combined conditions (Table 3). Alongside success and progress, we report DSQ (Eq. 3) to measure how well DA2 preserves terrain geometry under each perturbation.

The results show a clear two-tier structure (Figure 7). All conditions with surface texture—whether seen during training or not—maintain $DSQ > 0.85$ and degrade gracefully, with realistic textures like stone tiles and gravel within 5 pp of baseline. Brightness and color jitter cause ≤ 7 pp drops at moderate severity, and all lighting conditions preserve $DSQ > 0.91$. Featureless surfaces are the sole catastrophic failure mode: without visual features, DA2 cannot recover depth ($DSQ < 0.77$, success $< 23\%$). Since real-world surfaces universally carry natural texture, this failure mode is an artifact of simulation. DSQ thus serves as a deployment diagnostic: a pre-deployment check that $DSQ > 0.85$ rules out environments where the pipeline cannot function.

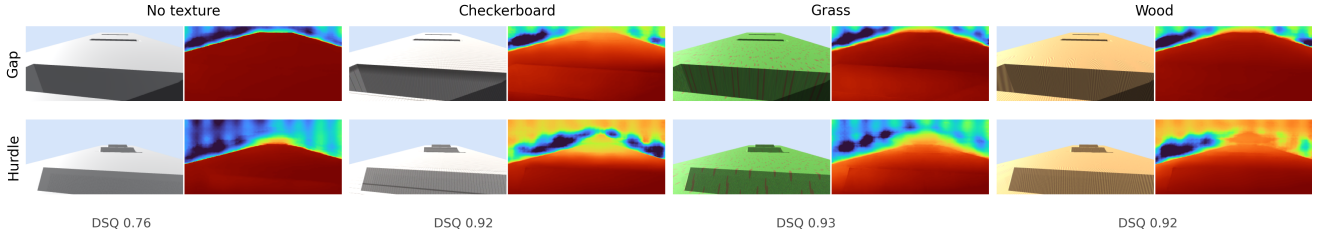


Figure 6: Robot-view comparison across representative terrain textures. Each panel shows rendered RGB (left) and the corresponding DA2 depth estimate (right).

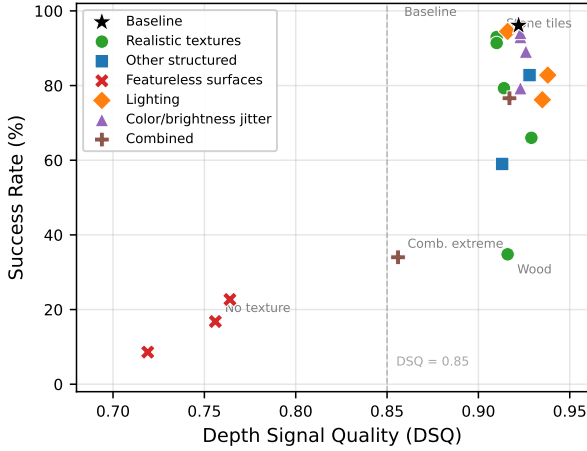


Figure 7: DSQ vs. success rate. Featureless surfaces (red) cluster below $DSQ = 0.85$ with catastrophic failure; all textured conditions maintain $DSQ > 0.85$.

6 Discussion and Conclusion

6.1 Summary

We presented a pipeline for vision-based quadruped parkour that uses monocular depth estimation as an intermediate representation. Our systematic experiments revealed that depth signal quality—governed by model scale and terrain texture—is the primary bottleneck for student performance, not network architecture. The best DA2-based student achieves 87% gap success, within 7 pp of the ground-truth depth upper bound.

The domain invariance experiment (Section 5.3) showed that DA2 generalizes to unseen textures and lighting conditions with graceful degradation, failing only on featureless surfaces that lack the visual features monocular depth estimation requires. The depth signal quality (DSQ) metric provides a practical deployment diagnostic, cleanly separating functional conditions from this failure mode.

6.2 Implications

The DA2 pipeline provides effective domain invariance for the visual perturbations most relevant to sim-to-real transfer: lighting variation and color shifts. This suggests that a student policy trained on DA2 depth in a simple simulation (with any structured texture) could transfer to real-world environments without photorealistic rendering, provided the real surfaces have natural visual texture—which is nearly universally the case.

The key practical implication is a separation of concerns: the depth estimator handles domain invariance (absorbing appearance variation), while the student policy handles behavior (mapping depth to actions). This modularity means improvements to either component are independent: better depth models improve signal quality without retraining the policy; better training curricula improve behavior without changing the visual pipeline.

6.3 Limitations

- **No real-world deployment.** All experiments are in simulation. Real-world validation is needed to confirm domain invariance transfers beyond simulated visual shifts.
- **Computational cost.** DA2-base (100M params) runs at ~ 40 ms/frame on an L40S GPU, which may be too slow for real-time deployment. Distillation into a smaller network or quantization could address this.
- **Texture requirement.** DA2 needs surface texture to produce useful depth estimates. Truly featureless surfaces (e.g., uniformly painted walls) may degrade performance. However, most real-world surfaces have natural texture.

6.4 Future Work

- **Real-world deployment** on the Unitree Go2 with onboard RGB camera.
- **Larger depth models:** DA2-large (335M params) or Video Depth Anything for temporal consistency.
- **End-to-end fine-tuning** of the depth estimator

Table 3: Domain invariance: DA2-base+texture student on gap Row 0 (256 episodes). Δ = change in success vs. baseline (pp). DSQ = depth signal quality (Eq. 3).

Perturbation	Succ.	Prog.	Δ	DSQ
Baseline (checkerboard)	96.1%	.945	—	.922
<i>Realistic surface textures</i>				
Concrete	79.3%	.856	−16.8	.914
Wood	34.8%	.563	−61.3	.916
Grass	66.0%	.766	−30.1	.929
Stone tiles	93.0%	.924	−3.1	.910
Gravel	91.4%	.918	−4.7	.910
<i>Other structured textures</i>				
Bricks	59.0%	.715	−37.1	.913
Noise	82.8%	.868	−13.3	.928
<i>Featureless surfaces</i>				
No texture	16.8%	.439	−79.3	.756
Solid red	22.7%	.492	−73.4	.764
Solid blue	8.6%	.377	−87.5	.719
<i>Lighting</i>				
Dim (0.3×)	82.8%	.855	−13.3	.938
Bright (2.0×)	94.5%	.934	−1.6	.916
Low contrast ($\gamma=2.2$)	76.2%	.817	−19.9	.935
<i>Color / brightness jitter</i>				
Brightness $\pm 30\%$	93.0%	.927	−3.1	.923
Brightness $\pm 60\%$	94.1%	.933	−1.6	.923
Color $\pm 15\%$	89.1%	.900	−7.0	.926
Color $\pm 30\%$	79.3%	.831	−16.8	.923
<i>Combined</i>				
Mild	76.6%	.833	−19.5	.917
Extreme	34.0%	.560	−62.1	.856

jointly with the student policy.

- **Multi-task evaluation** across diverse real-world environments (indoor, outdoor, varied lighting).

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